



Week 2 Workshop

Pre-processing Textual Data

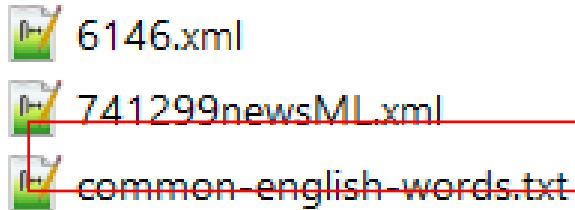
IFN647

Solutions

Task 1

Time: 30mins

Task 1: Write a program that loads (reads) an XML document, and prints out the **itemid** and the number of words in **<text>** of the document.



Output:

Document itemid: 741299 contains: 199 words

The following is the Format of the Document:

```
<newsitem xml:lang="en" date="1997-07-20" id="root" itemid="741299">
<title>BELGIUM: MOTOR RACING-LEHTO AND SOPER HOLD ON FOR GT
VICTORY.</title>
<headline>MOTOR RACING-LEHTO AND SOPER HOLD ON FOR GT
VICTORY.</headline>
<dateline>SPA FRANCORCHAMPS, Belgium</dateline>
<text>
<p>J.J. Lehto of Finland and Steve Soper of Britain drove their ailing
McLaren to victory in the fifth round of the world GT championship on
Sunday, beating the Mercedes of German Bernd Schneider and Austrian
Alexander Wurz by 15 seconds.</p>
<p>Their victory enabled them to open up a 16-point lead in the overall
standings over Schneider, who mounted a strong challenge on the
struggling leaders in the final minutes of the four-hour race.</p>
<p>But Soper, struggling with the car's handling caused by a broken
undertray, just managed to hold on for the win.</p>
<p>Lehto had opened up a lead of over 90 seconds during a mid-race
downpour in the Ardennes mountains.</p>
<p>"I thought that everyone else was driving on dry-weather tyres," he
joked afterwards.</p>
<p>"We swapped to rain tyres at exactly the right time and I was able
to push hard and open up a big lead."</p>
<p>Third to finish was the Porsche of France's Bob Wollek and Yannick
Dalmas and Belgian Thierry Boutsen.</p>
<p>The Belgian, a former Formula One driver, switched from the car he
normally shares with German Hans Stuck following a power-steering
failure on his own car.</p>
</text>
<copyright>(c) Reuters Limited 1997</copyright>
</newsitem>
```

```
IDLE Shell 3.11.7
File Edit Shell Debug Options Window Help
Python 3.11.7 (tags/v3.11.7:fa7a6f2, Dec 4 2023, 19:24:49) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\N11016647\OneDrive - Queensland University of Technology\Desktop\IFN647_2025\wk2_workshop_solutions\wk2_workshop_solutions\wk2_task1_IFN647.py
Document itemid: 741299 contains: 199 words
>>>
```

Workshop Task 1 - Hint

How to only read part of the text?

If find `<text>`:

read data

If find `</text>`:

stop reading

for line in file.readlines():

if line.startswith(' <text> '):

Check words number

if line.startswith(' </text> '):

break

line.strip()

line.split()

get rid of `<p>`, `</p>` and punctuations, and numbers.

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VICTORY.</title>
<headline>MOTOR RACING-LEHTO AND SOPER HOLD ON FOR GT
VICTORY.</headline>
<dateline>SPA FRANCORCHAMPS, Belgium</dateline>
```

<text>

<p>J.J. Lehto of Finland and Steve Soper of Britain drove their ailing McLaren to victory in the fifth round of the world GT championship on Sunday, beating the Mercedes of German Bernd Schneider and Austrian Alexander Wurz by 15 seconds.</p>

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<p>The Belgian, a former Formula One driver, switched from the car he normally shares with German Hans Stuck following a power-steering failure on his own car.</p>

</text>

<copyright>(c) Reuters Limited 1997</copyright>

</newsitem>

```
text = "Hello, world! How are you?"
words = text.split() # Default splits by whitespace
print(words)
```

```
data = "apple,banana,grape,orange"
fruits = data.split(",")
print(fruits)
```

```
text = "Python-is-awesome"
words = text.split("-")
print(words)
```

Solution (Task 1) - Explain

```
import string
```

```
#myfile=open('C:\\Teaching\\Units\\IFN647\\workshops\\wk2_solutions\\data\\741299newsML.xml', 'r')  
myfile=open('741299newsML.xml', 'r') # data in the current folder
```

```
file_=myfile.readlines()  
word_count = 0  
for line in file_:  
    line = line.strip()
```

```
myfile.close()  
print('Document itemid: '+ docid+ ' contains: '+ str(word_count) + ' words')
```

Solution (Task 1) - Explain

```
import string

#myfile=open('C:\\Teaching\\Units\\IFN647\\workshops\\wk2_solutions\\data\\741299newsML.xml', 'r')
myfile=open('741299newsML.xml', 'r') # data in the current folder

start_end = False
file_=myfile.readlines()
word_count = 0
for line in file_:
    line = line.strip()
    if(start_end == False):

elif line.startswith("</text>"):

else:

myfile.close()
print('Document itemid: '+ docid+ ' contains: '+ str(word_count) + ' words')
```

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#myfile=open('C:\\Teaching\\Units\\IFN647\\workshops\\wk2_solutions\\data\\741299newsML.xml', 'r')
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myfile=open('741299newsML.xml', 'r') # data in the current folder

start_end = False
file_=myfile.readlines()
word_count = 0
for line in file_:
    line = line.strip()
    if(start_end == False):
        if line.startswith("<newsitem "):
            for part in line.split():
                if part.startswith("itemid="):
                    docid = part.split("=")[1].split("\\")[1]
                    break
            if line.startswith("<text>"):
                start_end = True
    elif line.startswith("</text>"):
        break
    else:

myfile.close()
print('Document itemid: ' + docid+ ' contains: ' + str(word_count) + ' words')
```

Solution (Task 1) - Explain

```
import string
```

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#myfile=open('C:\\Teaching\\Units\\IFN647\\workshops\\wk2_solutions\\data\\741299newsML.xml', 'r')
myfile=open('741299newsML.xml', 'r') # data in the current folder
```

```
start_end = False
file_=myfile.readlines()
word_count = 0
for line in file_:
    line = line.strip()
    if(start_end == False):
        if line.startswith("<newsitem "):
            for part in line.split():
                if part.startswith("itemid="):
                    docid = part.split("=")[1].split("\\")[1]
                    break
            if line.startswith("<text>"):
                start_end = True
        elif line.startswith("</text>"):
            break
    else:
        line = line.replace("<p>", "").replace("</p>", "")
        line = line.translate(str.maketrans('', '', string.digits)).translate(str.maketrans(string.punctuation, ' '*len(string.punctuation)))
        for term in line.split():
            word_count += 1
myfile.close()
print('Document itemid: '+ docid+ ' contains: '+ str(word_count) + ' words')
```

This removes `<p>` and `</p>` tags from `line`.

If `line` contains `"<p>Hello</p>"`, it becomes `"Hello"`.

The first two arguments (`' '`, `' '`) are empty because we are not replacing any characters.

The third argument (`string.digits`) specifies the characters to be deleted (i.e., `"0123456789"`).

This tells `translate()` to remove all digits.

`str.maketrans(string.punctuation, ' '*len(string.punctuation))` creates a translation table that replaces all punctuation with spaces.

Example:

- `"Hello, World!"` → `"Hello World "`



Task 2 Solution

```
import glob, os
import string
|
def parse_doc(inputpath, stop_ws):
    coll = {}
    os.chdir(inputpath)
    # Example of "inputpath" when you Run...Customized in Mac:
    # /Users/li/IFN647/wk2_workshop_solutions

    myfile=open('6146.xml')

    curr_doc = {}
    start_end = False

    file_=myfile.readlines()
    word_count = 0 #wk2
    for line in file_:
        line = line.strip()
        #print(line)
        if(start_end == False):
            if line.startswith("<newsitem "):
                for part in line.split():
                    if part.startswith("itemid="):
                        docid = part.split("=")[1].split("\")[1]
                        break
            if line.startswith("<text>"):
                start_end = True
        elif line.startswith("</text>"):
            break
        else:
            line = line.replace("<p>", "").replace("</p>", "")
            line = line.translate(str.maketrans('', '', string.digits)).translate(str.maketrans(string.punctuation, ' '*len(string.punctuation)))
            line = line.replace("\\s+", " ")
            for term in line.split():
                word_count += 1 #wk2
                term = term.lower() #wk2
                if len(term) > 2 and term not in stop_words: #wk2
                    try:
                        curr_doc[term] += 1
                    except KeyError:
                        curr_doc[term] = 1

    myfile.close()
    return(word_count, {docid:curr_doc})
    # return a tuple, the first element is the number of words in <text> and
    # the second one is a dirctionary that includes only one pair of doc_id and a disctionary of term_frequency pairs
```

←find the doc ID

←process each paragraph

←calculate term frequency

Task solution Highlights: stop words

Stopping

- **Stop words** are the words in a **stop list** (or **stoplist** or *negative dictionary*) that are filtered out (i.e. stopped) before or after processing of natural language data (text) because they are insignificant [1].

common-english-words.txt

```
a,able,about,across,after,all,almost,also,am,among,an,and,any,are,as,at,be,because,been,but,by,can,cannot,could,dear,did,do,does,either,else,ever,every,for,from,get,got,had,has,have,he,her,hers,him,his,how,however,i,if,in,into,is,it,it's,just,least,let,like,likely,may,me,might,must,my,neither,no,nor,not,of,off,often,on,only,or,other,our,own,rather,said,say,says,she,should,since,so,some,than,that,the,their,them,then,there,these,they,this,tis,to,too,twas,us,wants,was,we,were,what,when,where,which,while,who,whom,why,will,with,would,yet,you,your
```

Task 3 Solution

```
if __name__ == '__main__':
```

← This ensures that the script runs only when executed directly (e.g., python script.py in the terminal) , not when imported as a module. (e.g import my_script in another script)

```
import sys
```

```
if len(sys.argv) != 2:
```

```
    sys.stderr.write("USAGE: %s <coll-file>\n" % sys.argv[0])
```

```
    sys.exit()
```

```
stopwords_f = open('common-english-words.txt', 'r') # wk2
```

```
stop_words = stopwords_f.read().split(',')
```

```
stopwords_f.close()
```

```
# call the function by using Run...Customized
```

```
x = parse_doc(sys.argv[1], stop_words) # you need to input the real "inputpath"
```

```
print('--- Task2: The return value of function parse_doc ---')
```

```
print(x)
```

```
print('--- The outcomes of Task3 ---')
```

```
for doc in x[1].items():
```

```
    print('Document itemid: ' + doc[0] + ' contains: ' + str(x[0]) + ' words and ' + str(len(doc[1])) + ' terms')
```

- sys.argv is a **list** in Python that stores command-line arguments.
- sys.argv[0] is always the script name.
- sys.argv[1] is the first argument passed to the script.
- len(sys.argv) != 2 checks whether the user **did not provide exactly one argument**.